

Gautam Gururaj Molakalmuru

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Summary

Software Engineer with 3+ years of experience designing and operating scalable backend APIs and services in cloud environments using Java, Python, and Spring Boot. I have helped build products from pilot to global launch, engineered AI/ML and GenAI-enabled workflows, and published research on agentic AI in MLOps. I bring hands-on experience with LLM integration, RAG pipelines.

Education

The Ohio State University	Columbus, USA
Master of Science in Computer Science and Engineering - GPA: 3.85/4	May 2026
New Horizon College of Engineering	Bangalore, India
Bachelor of Engineering in Computer Science - GPA: 8.85/10	August 2022

Technical Skills

Languages: Java, Reactive Java, Python, JavaScript.
Frameworks & Architecture: Spring Boot, ReactJS, Microservices, RESTful APIs, Event-Driven Architecture.
Data & Messaging: Apache Kafka, MongoDB, Elasticsearch, Redis, PostgreSQL, MySQL.
Cloud & Infra: AWS (Certified), Kubernetes, Docker, Linux, Jenkins (CI/CD), GCP.
ML & Computer Vision: YOLO, OpenCV, PyTorch, scikit-learn, Pandas; model retraining, MLOps, LLM integration, RAG pipelines.
Testing & Quality: JUnit, Git, SonarQube; automated testing, CI/CD, monitoring, alerting, logging.

Professional Experience

Texas Advanced Computing Center (UT Austin)	Austin, Texas
Software Development Intern	Jun 2025 - Aug 2025
- Enhanced Python plugins for image detection, scoring, and cloud publishing, deploying them to edge devices for real-time edge computing. - Optimized the inference pipeline to slash end-to-end processing from minutes to less than 5 seconds by optimizing data flow and custom runtime configurations. - Designed a motion-triggered detection system using YOLO for real-time inference, fine-tuning the model to boost accuracy from 50% to 95%. - Implemented an event-driven architecture to publish detection events and built downstream consumer services to establish a complete end-to-end workflow, with monitoring and logging across each stage for operational visibility.	
Quinbay (Blibli.com)	Bangalore, India
Software Engineer	July 2022 - July 2024
- Played a key role in engineering scalable backend microservices using Java and Spring Boot, driving features through the full SDLC from pilot to global production launch. - Optimized performance using Reactive Java and parallel processing, boosting throughput by 70% and enabling high concurrency. - Implemented a catalog service to ingest and process millions of data points using Kafka, and managed Elasticsearch and MongoDB for scalable storage and fast retrieval. Milestone: Joined the pilot project at inception and helped drive the product to its first successful launch within 10 months. Business impact: Led onboarding and technical support for business users, resolving post-launch defects and stabilizing the platform, cutting critical incidents by 50%. Leadership: Mentored 3+ junior engineers through weekly sessions and code reviews, boosting throughput by 40% in one quarter.	

Academic Projects

ICICLE Intelligence Plane Graduate Research Associate (Part-Time)	Jan 2025 - Present
- Built scalable cloud orchestration workflows using Python and Kafka, integrating LLM-powered agentic components to automate the end-to-end ML lifecycle and cut execution time from 12 hours to under one hour. - Architected a closed-loop GenAI-enabled MLOps platform as shared internal tooling, automating data ingestion, model retraining, and continuous delivery, with observability, alerting, and operational guardrails built into the pipeline.	

Achievements

- Successfully earned the AWS Certified Cloud Practitioner certification (view badge). - Received recognition from Quinbay for contributions during product launch and post-launch phases. - Published a paper titled Beyond Automation: Integrating Agentic Capabilities into MLOps, accepted at PEARC '25 conference. - Presented a poster and delivered a lightning talk on Reimagining the ICICLE Intelligence Plane: A Framework for MLApps Lifecycle Management at the ICICLE All-Hands Meeting. - Presented a poster titled Operational Considerations for Real-Time ML Pipelines on Edge devices, at Gateways '25 conference. - Selected and completed a semester-long, merit-based study abroad exchange program at ESIGELEC University, France.	
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